

WORKSHEET A.5.M3

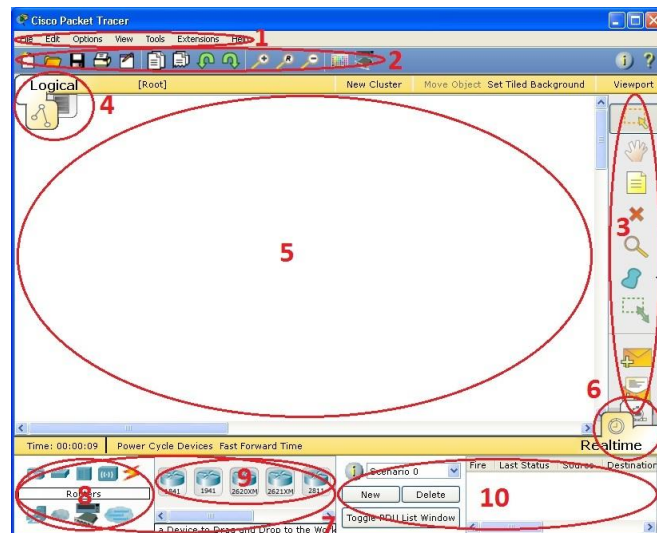
Creating a Simple Network

What we will learn today:

- Build a network.
- Identify problems in small networks and propose and apply solutions.

Consolidation Activities

1. Start Cisco Packet Tracer.

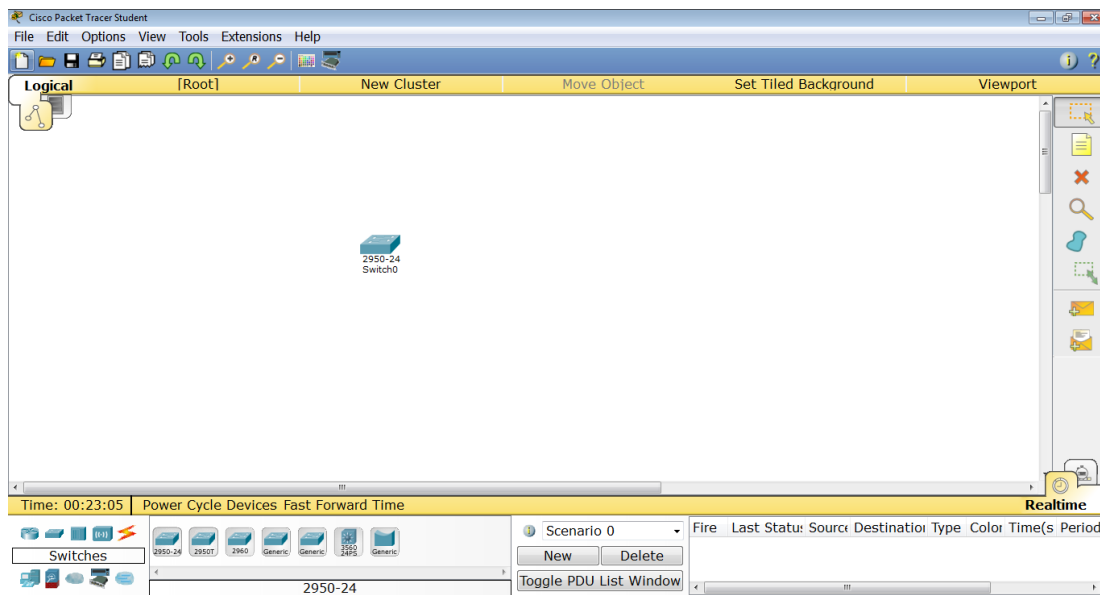


The initial interface contains ten main components. If you are unsure what an item does, move the pointer over it to display help. The most important components are 1, 3, 5 and 7.

Brief description:

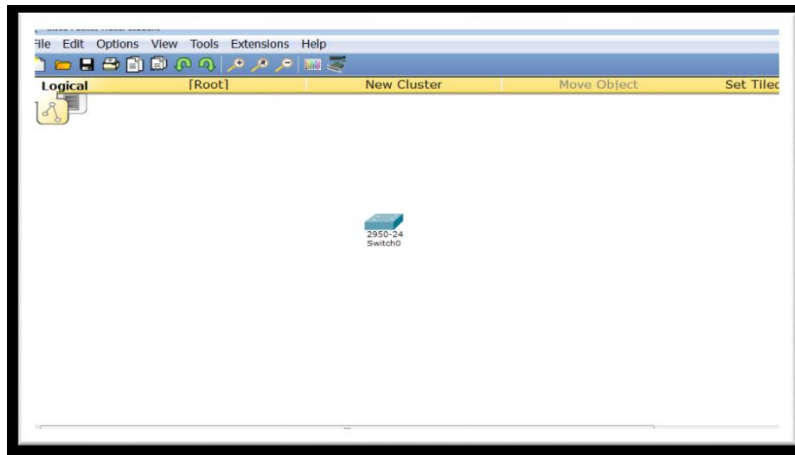
1. Menu Bar: File, Edit, Options, View, Tools, Extensions and Help. It contains commands such as Open, Save, Print and Preferences. Help includes videos on using Packet Tracer.
2. Main Tool Bar: Copy, Paste, Undo, Redo, Zoom, Drawing Palette and Devices Dialog.
3. Common Tools Bar: Select, Move Layout, Place Note, Delete, Inspect, Resize Shape, Add Simple PDU and Add Complex PDU.
4. Logical/Physical Workspace: switches between logical and physical views.
5. Workspace: create the network, run simulations, and view information and statistics.

6. Realtime/Simulation Bar: switches between real-time and simulation modes.
 7. Network Component Box (8 and 9): choose devices and connections for the diagram.
 8. Device-Type Selection Box: choose a device or connection category. The neighbouring box changes accordingly.
 9. Device-Specific Selection Box: choose a specific device model.
 10. User-Created Packet Window: monitor test packets.
2. Design your first network using:
 - (a) One switch, which can connect two or more computers at the same time.
 - (b) Two computers.
 - (c) Cables.
 3. Drag a switch to the centre of the workspace as shown below.

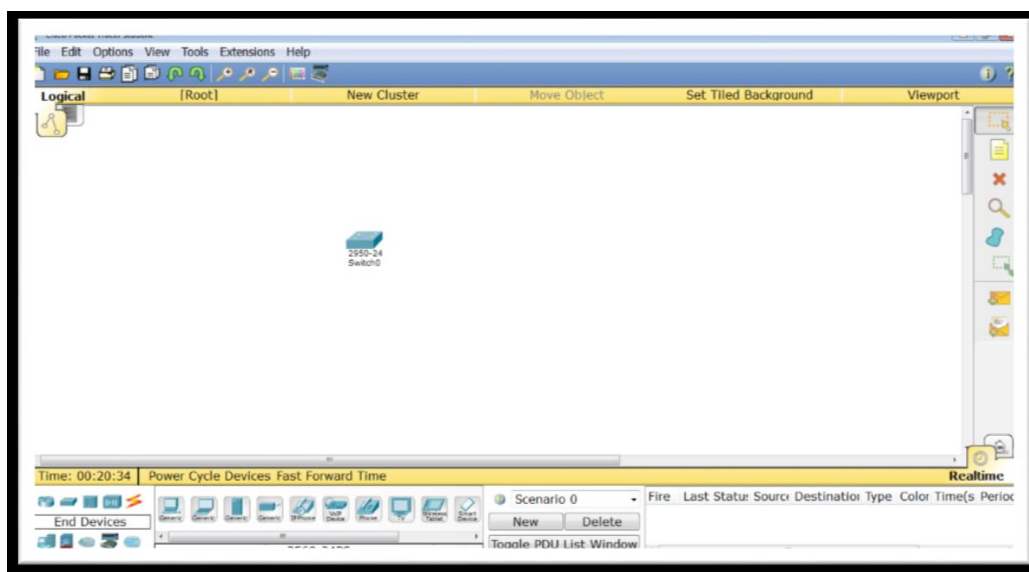


Select the Switches device type, then select the 2950-24 model. The number 24 means that up to 24 computers can be connected.

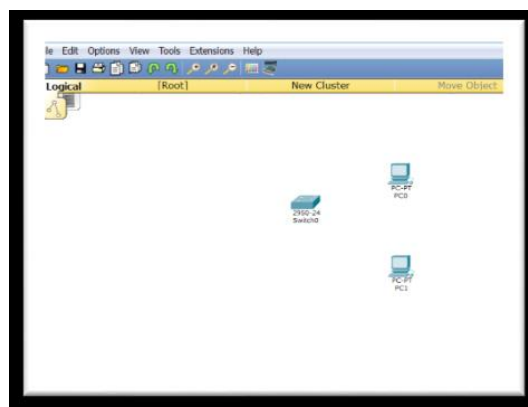
The result should resemble the following figure:



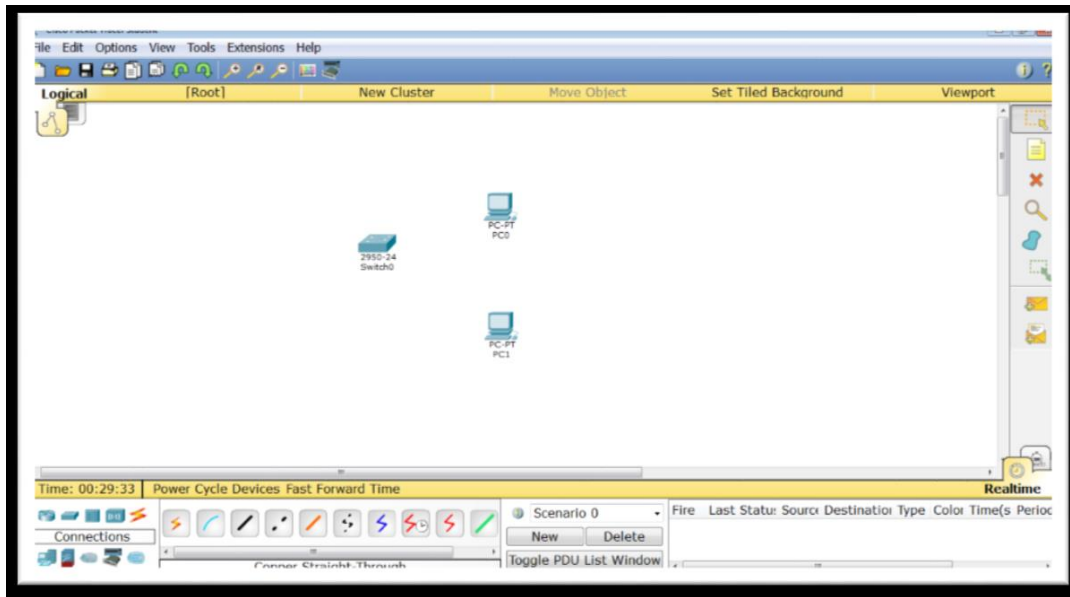
4. Select an End Device of type PC and drag it to the right of the switch. Repeat to add a second PC.



Choose End Devices, then the Generic PC model. The result should resemble:

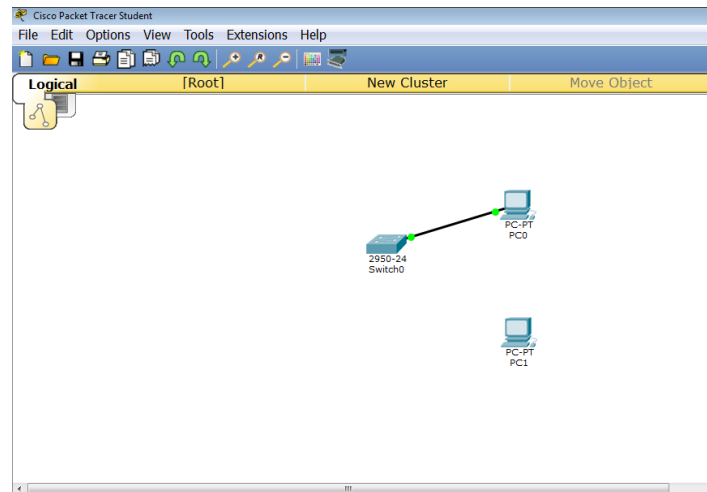


5. Connect the devices using cables as shown below.



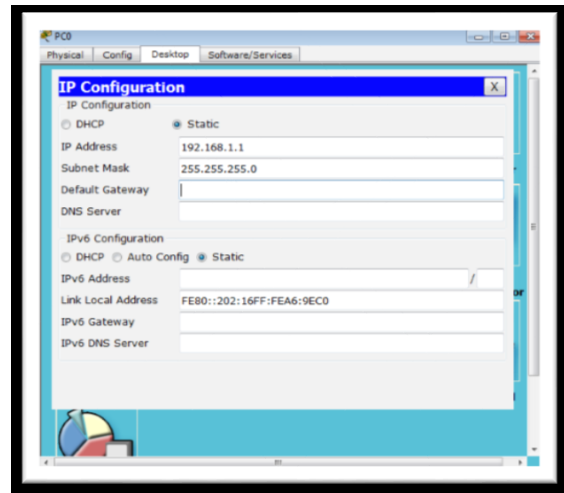
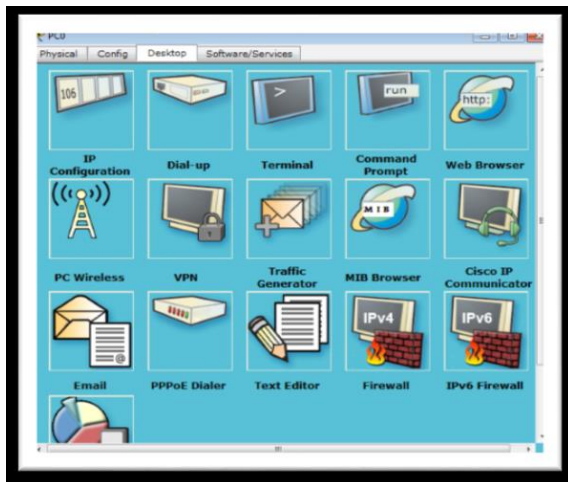
Select Connections, then Copper Straight-Through.

- (a) Select the switch and choose FastEthernet0/1; this is the first port on the switch.
- (b) Select PC0 and choose FastEthernet0.
- (c) If the steps were completed correctly, the network should resemble the figure below. If the lights are orange, wait 2-3 seconds. If they are red, the wrong cable or port was used.
- (d) Connect the second PC in the same way.



6. After making the physical connections, assign logical addresses (IP addresses).

- (a) Select PC0, then the Desktop tab.
- (b) Select IP Configuration and enter:
IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0
- (c) Close both windows.



7. Configure PC1:

IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0

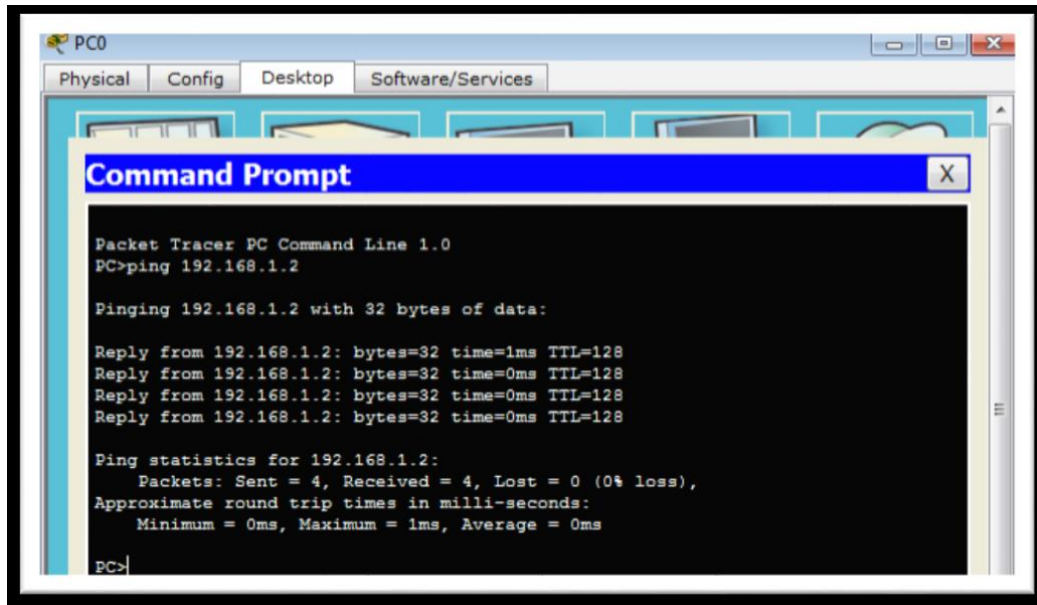
The IP address must be different for every device connected to the same network.

8. Save the file using File > Save (or the disk icon) as Network1 in the folder specified by your teacher.

9. Your network is ready. Test it using both methods below:

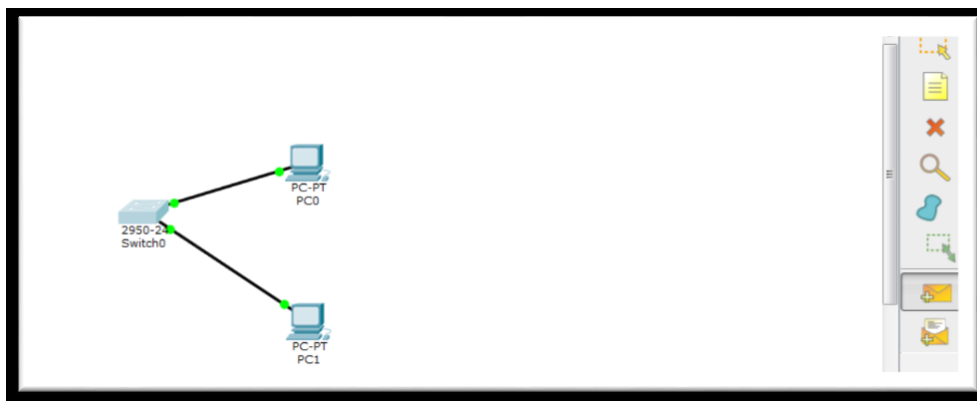
(a) Using the command line and ping:

- Select PC0 > Desktop > Command Prompt.
- Type: ping 192.168.1.2

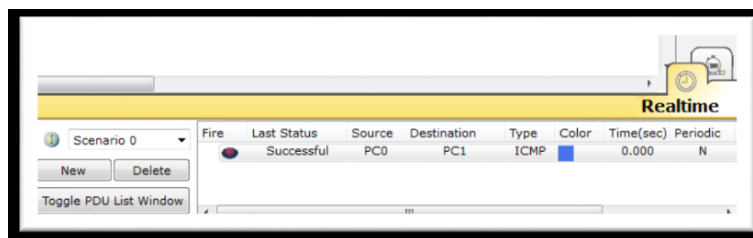


A successful test sends four data packets and receives four replies.

(b) Using the simulator:



Select Add Simple PDU, then select PC0 and PC1 in sequence. If Successful appears in the lower-right corner, the network is working.

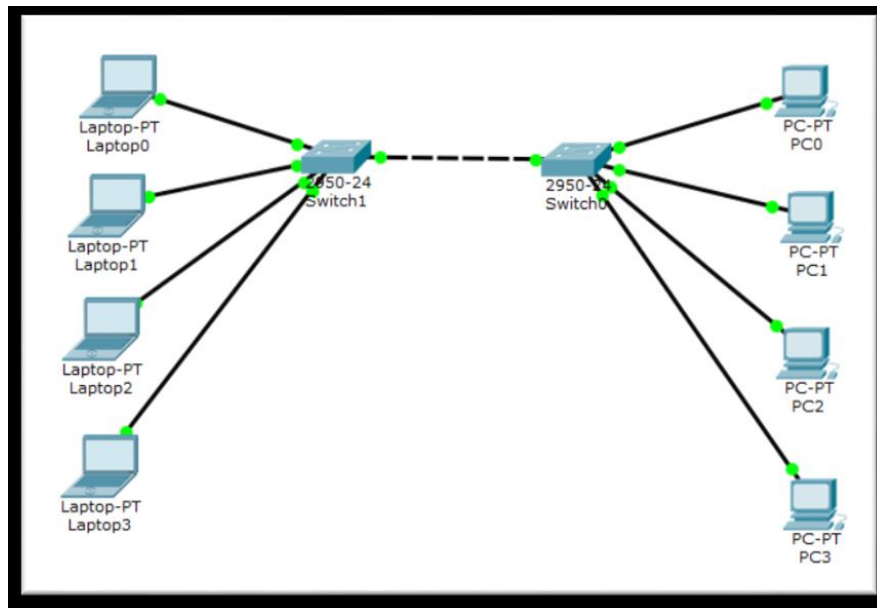


Additional Activities

10. Connect two more PCs and configure the appropriate Copper Straight-Through cables and IP addresses so that the network works.

11. Add another switch and connect it using a Copper Cross-Over cable. Connect four laptops and assign suitable IP addresses.

The result should resemble:



12. Open Troubleshooting 1.pka and find the errors in the network configuration.

13. With your teacher, implement this activity using a real small switch, laboratory computers and straight Ethernet cables.